Ministry of Foreign Affairs of the Netherlands

Food and Agriculture Organization of the United Nations

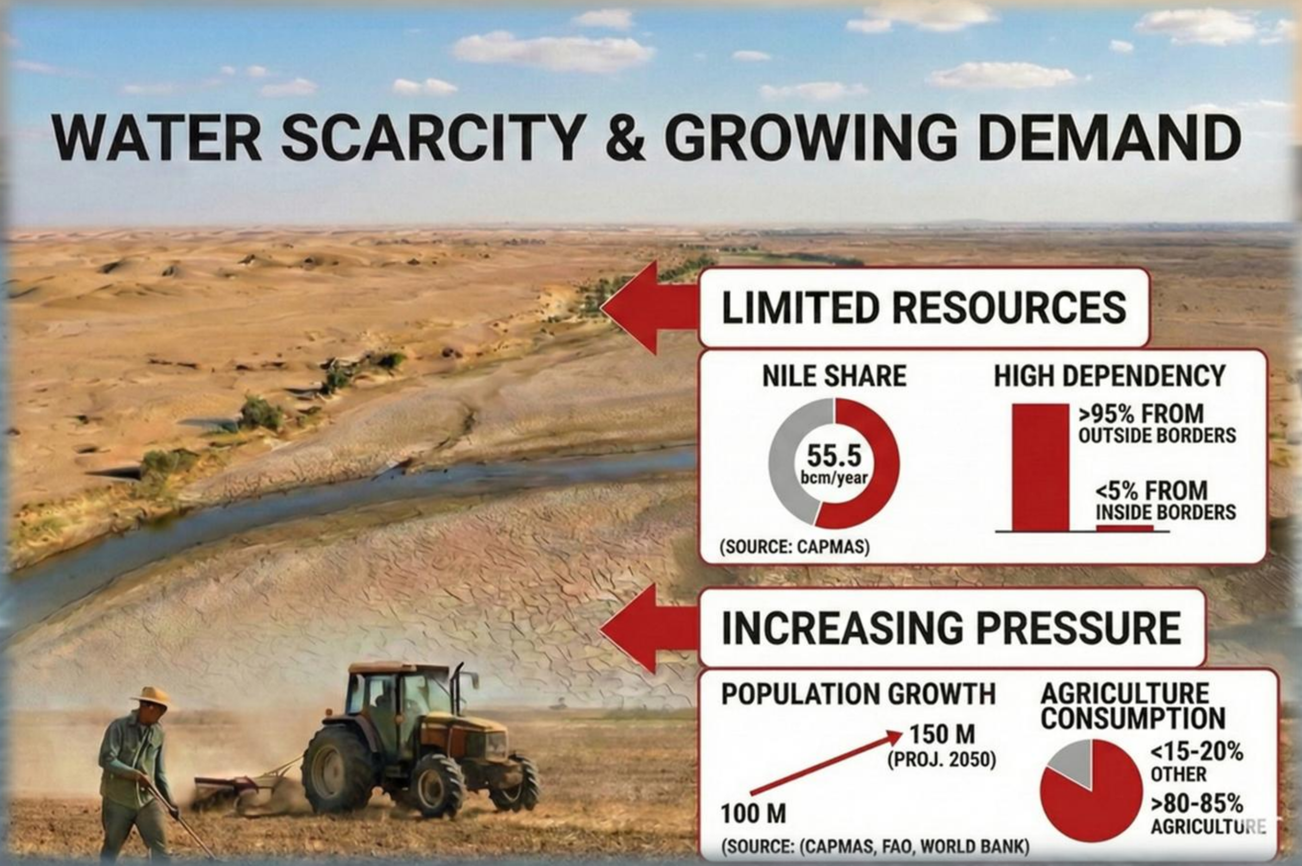
IHE DELFT
Institute for Water Education under the auspices of UNESCO

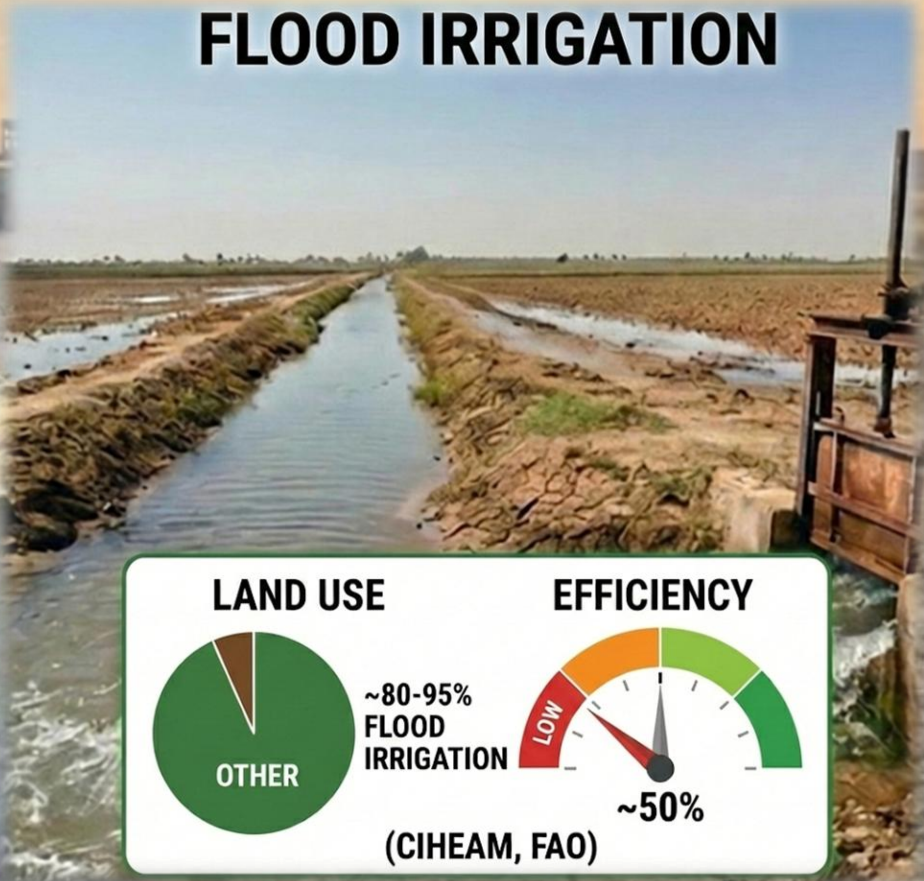
‘PRECISION IRRIGATION ADVISOR’

Smart Water Management for Egypt, Powered by WaPOR

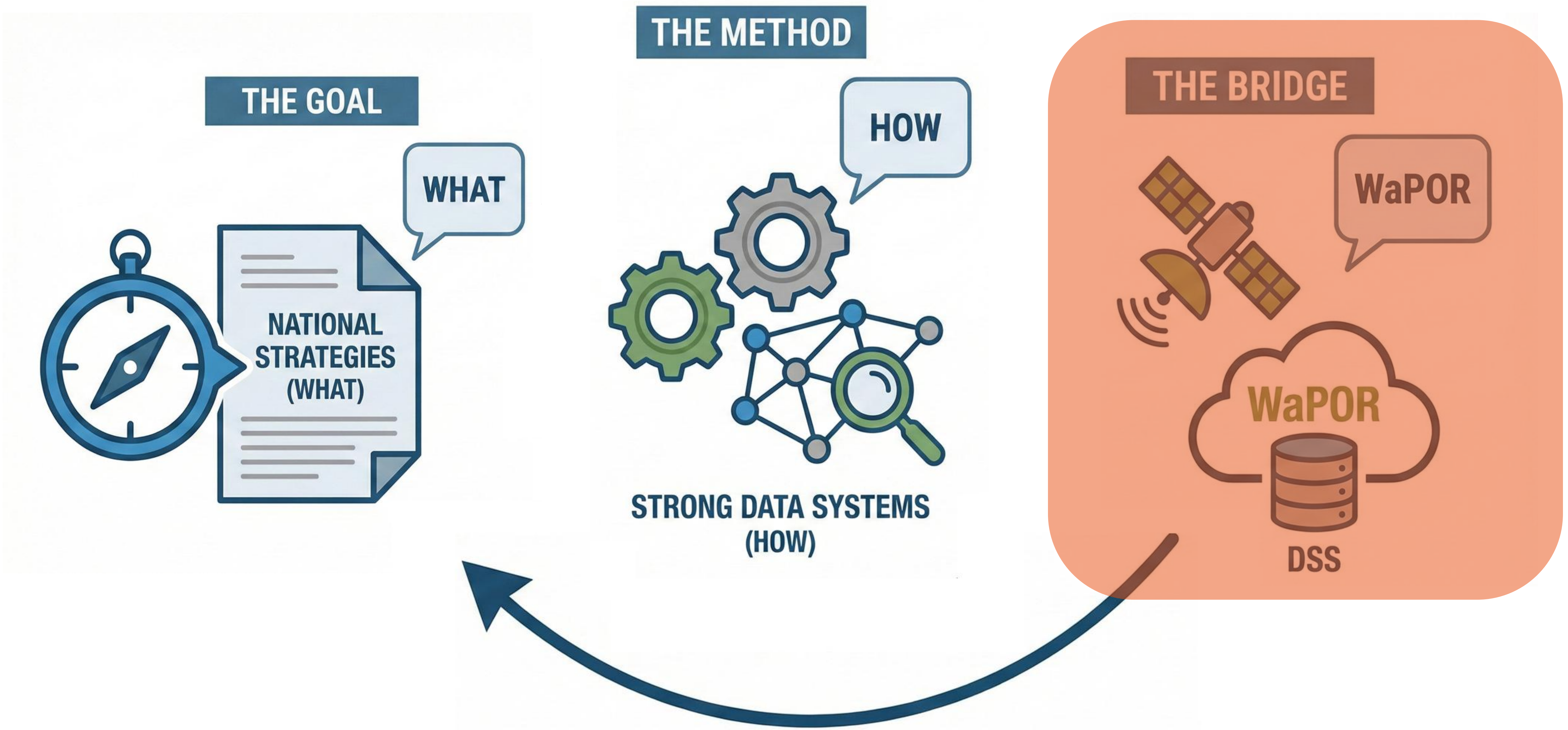
Ahmed Elnaggar (IHE Delft)

THE CHALLENGE





Source: Egyptian Ministry of Water Resources and Irrigation


THE GOAL
WHAT
NATIONAL STRATEGIES (WHAT)
THE METHOD
HOW
STRONG DATA SYSTEMS (HOW)
THE BRIDGE
WaPOR
WaPOR
DSS


WaPOR
Search for locations
Explore Data
Upload
Map
Actual evapotranspiration and interception
Time:
2023-09-11
No data
https://data.apps.fao.org/wapor/?lang=en
Analysis
Story
Map
Share / Print
English
Help
Login
FAO & IHE & IWMI
Remote Sensing Platform
Worldwide Accessibility
WaPOR
Water Productivity
Near Real-Time Info
Give Feedback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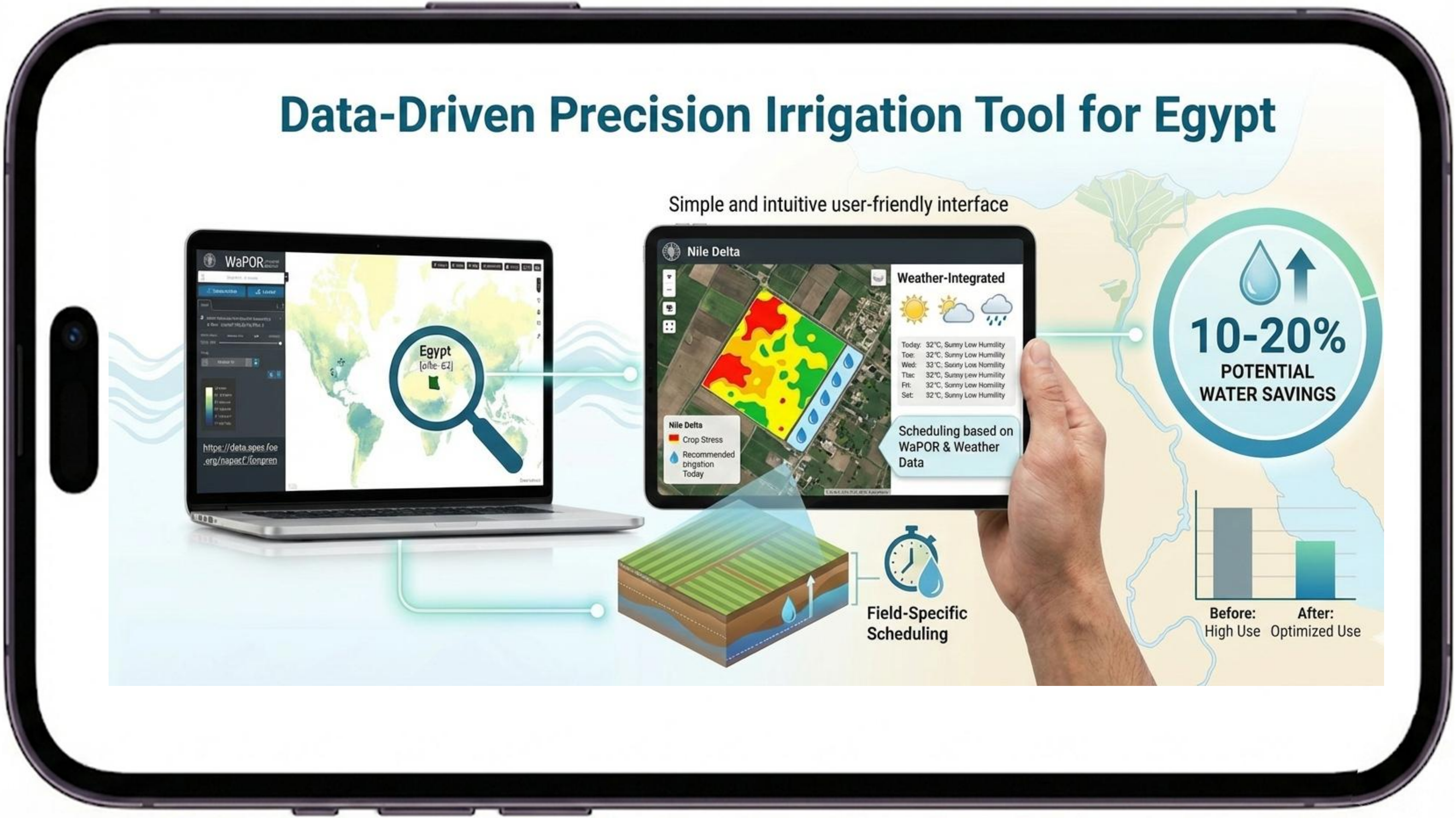
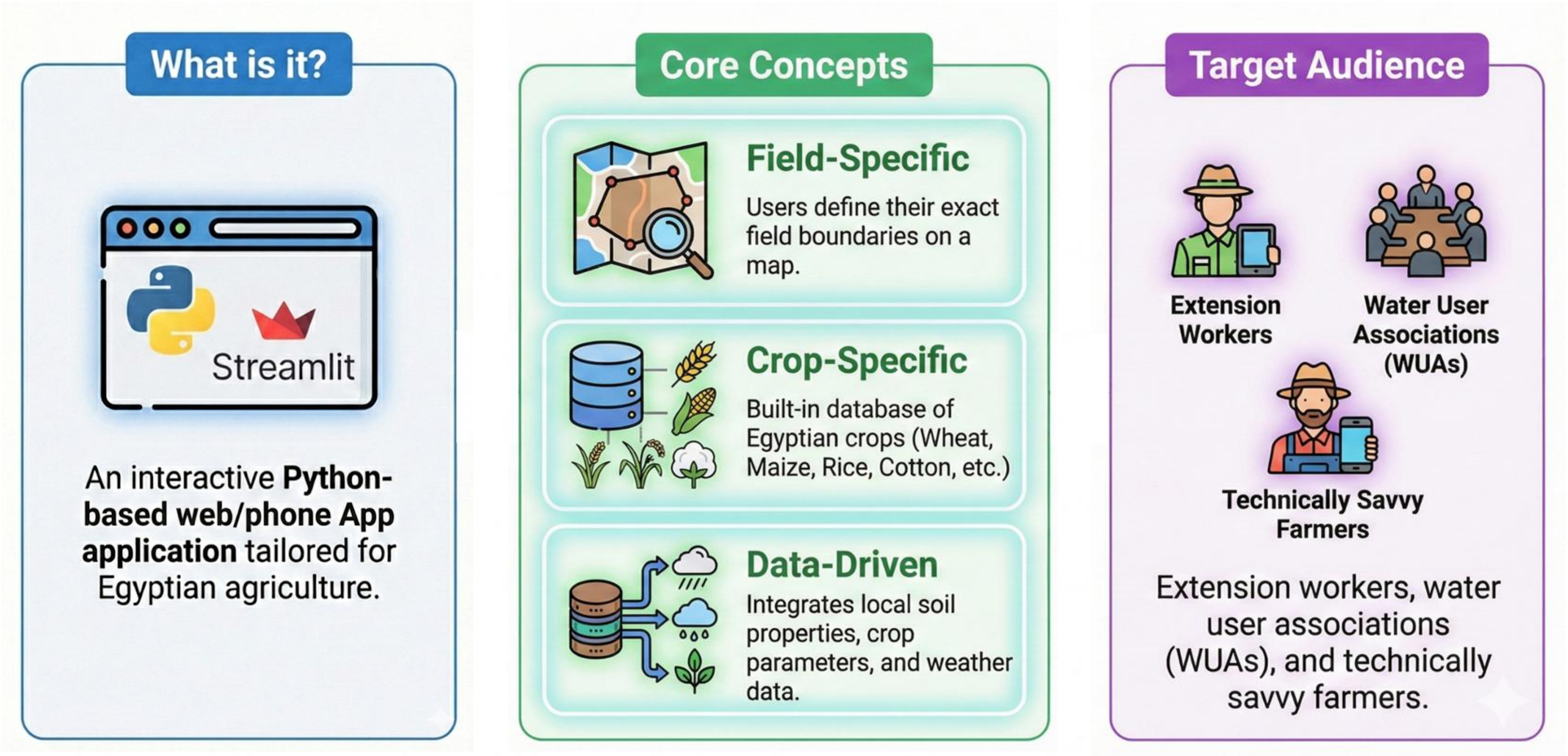
Contact
Innovation Meets Affordability
View case studies & data
Learn More
AI Applications in Agriculture
Advanced technology for farming
Democratized Precision Agriculture
Explore tools & sensors
Learn More
Low-Cost Smart Water Solutions
Affordable tools for water management
The democratized program promotes accessible precision agriculture.
Sustainable tools that support environmental stewardship through low-cost conditions and scalable solutions.
Develop data-driven sensors and precision-agriculture technologies.
Interactive Map
Project Data
Partners

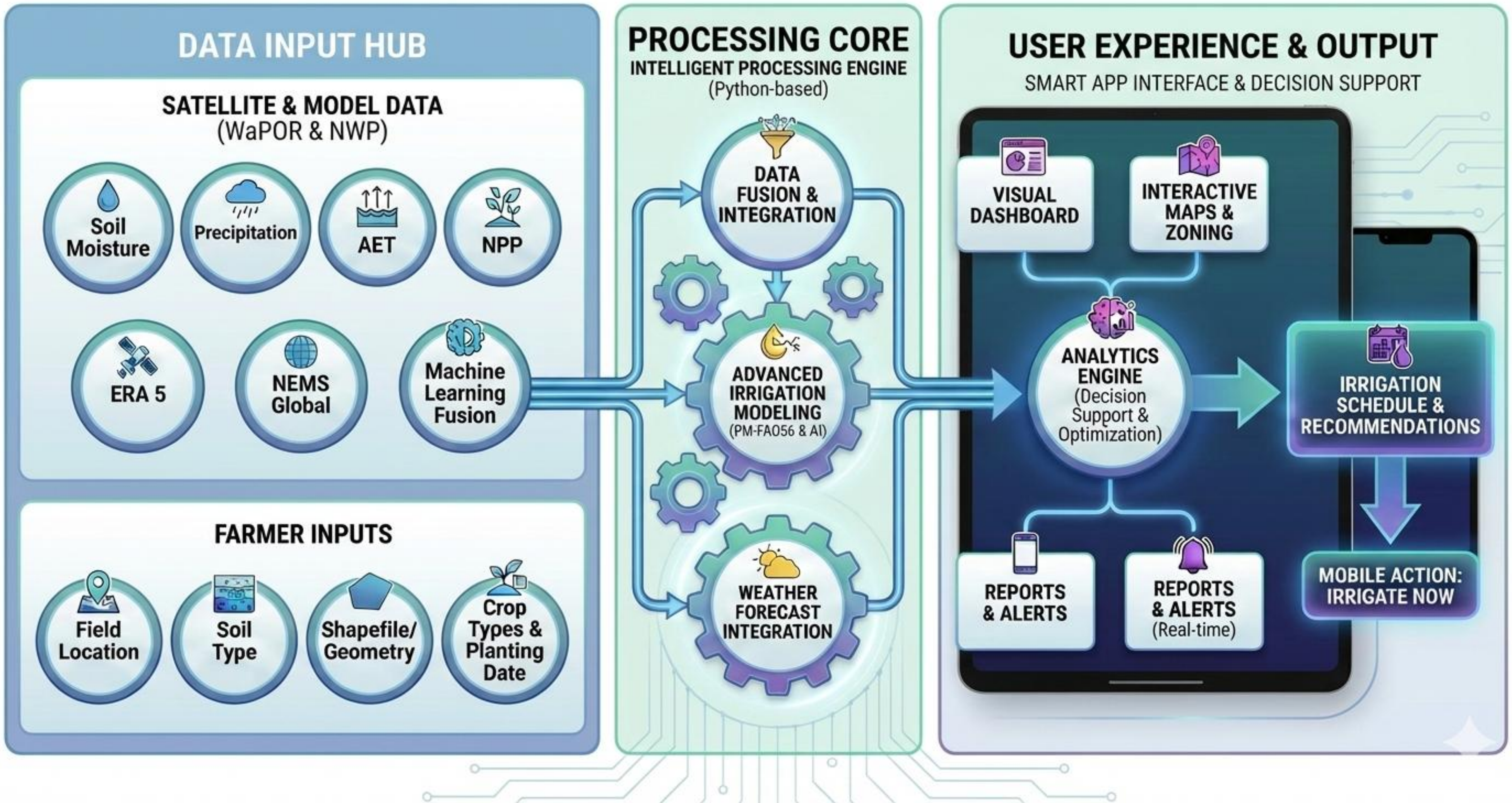


Data-Driven Precision Irrigation Tool for Egypt
Simple and intuitive user-friendly interface
WaPOR
Egypt
Nile Delta
Weather-Integrated
Nile Delta
Crop Stress
Scheduling based on WaPOR & Weather Data
10-20%
POTENTIAL WATER SAVINGS
Field-Specific Scheduling
Before: High Use
After: Optimized Use

Precision Irrigation Advisor



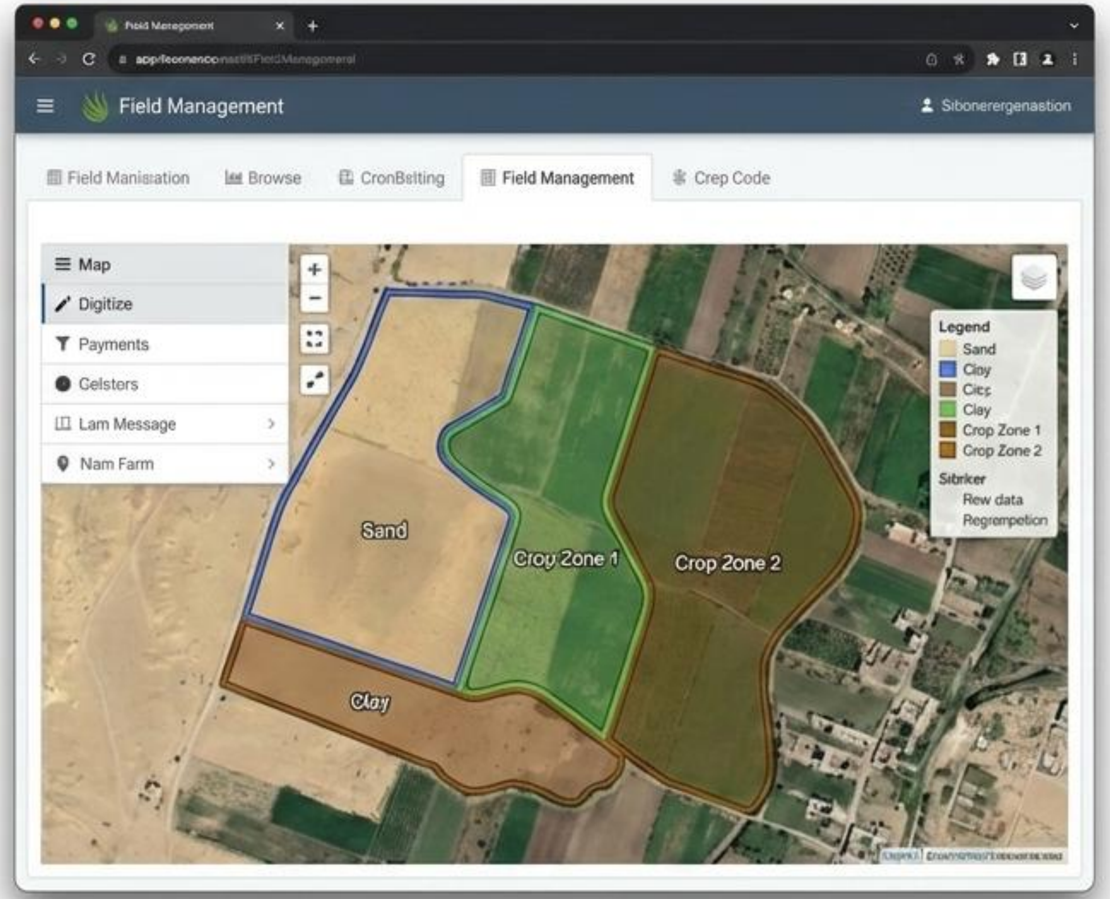
System Architecture



Key Features - Spatial Management

Powered by Folium & Geemap

Digitize Your Farm (Soil & Crop Zones)

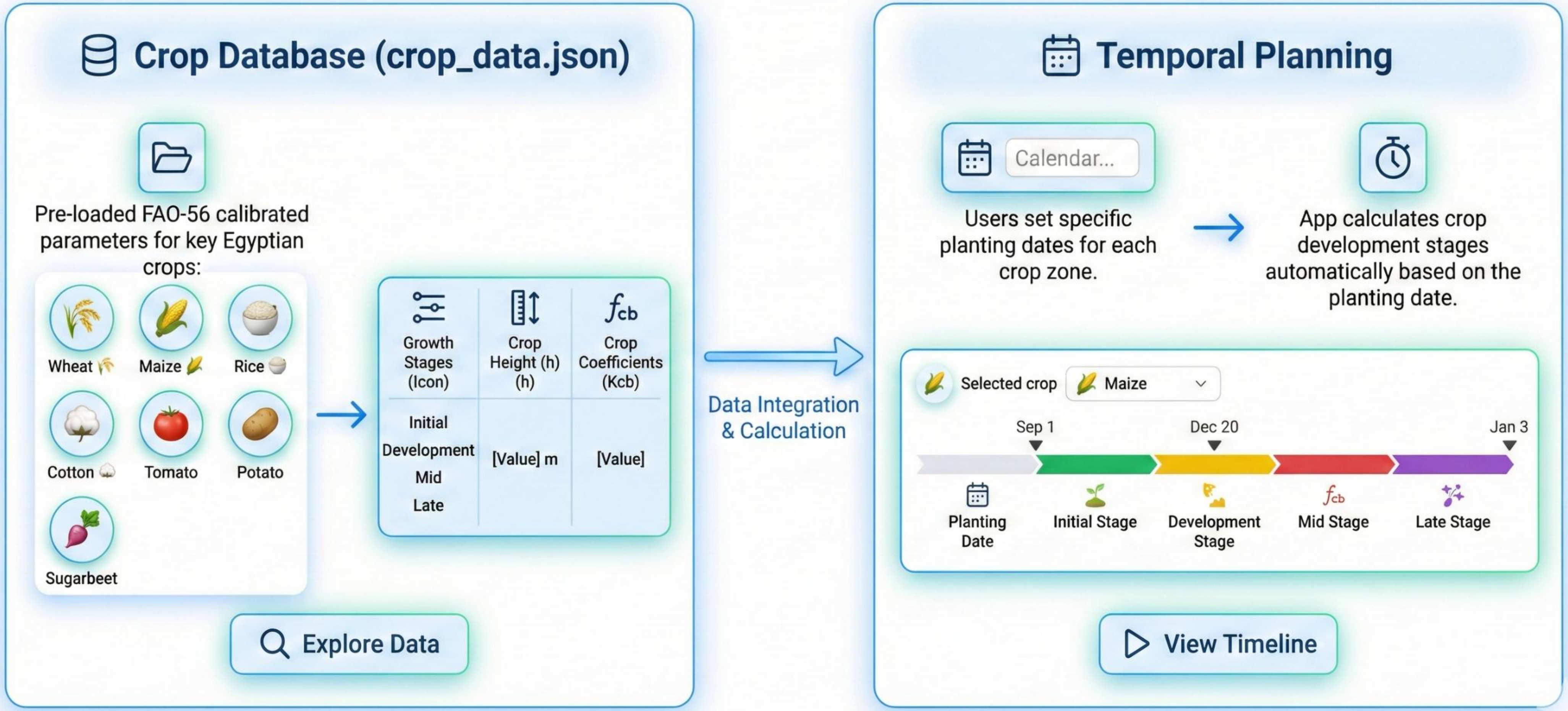


Soil Heterogeneity & Types

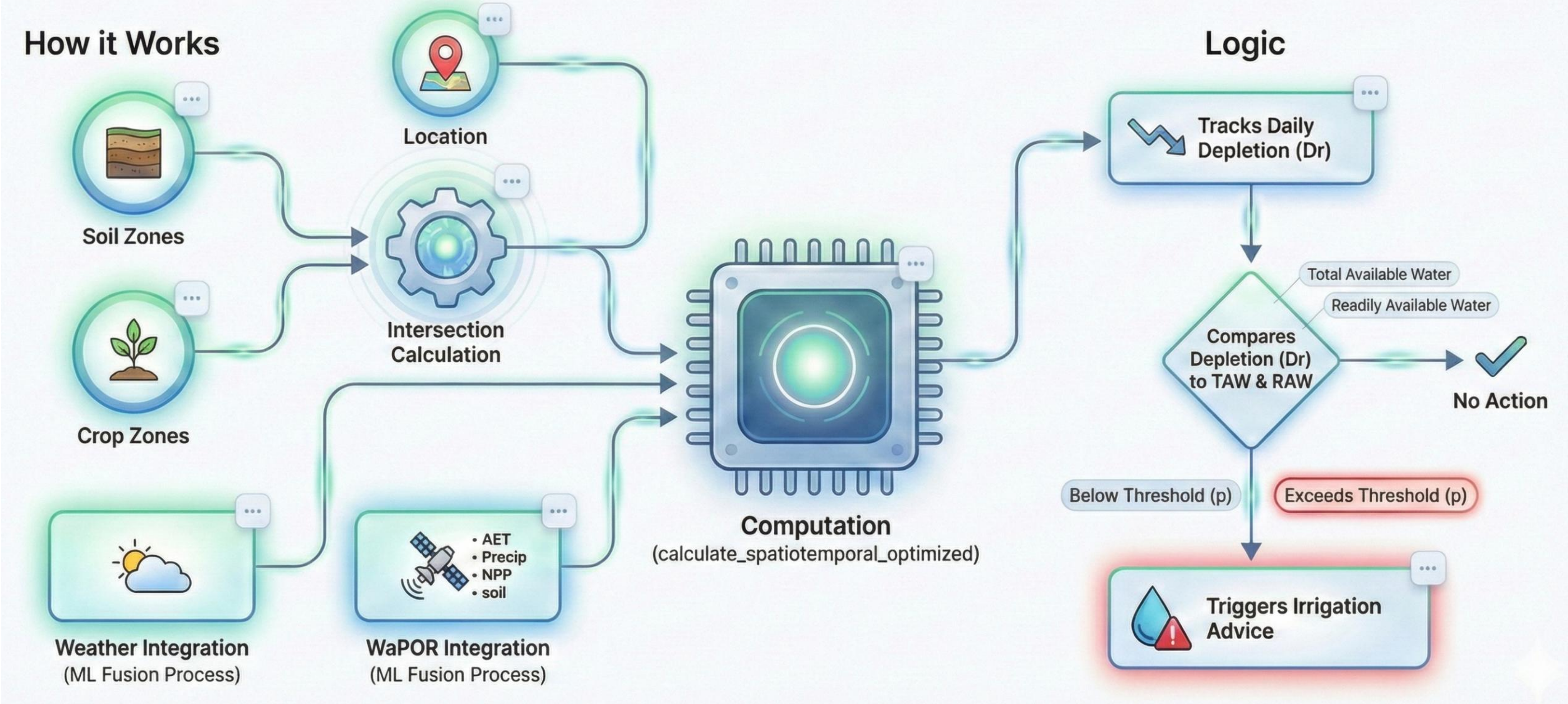
REW, TEW, TAW

Customizable Properties

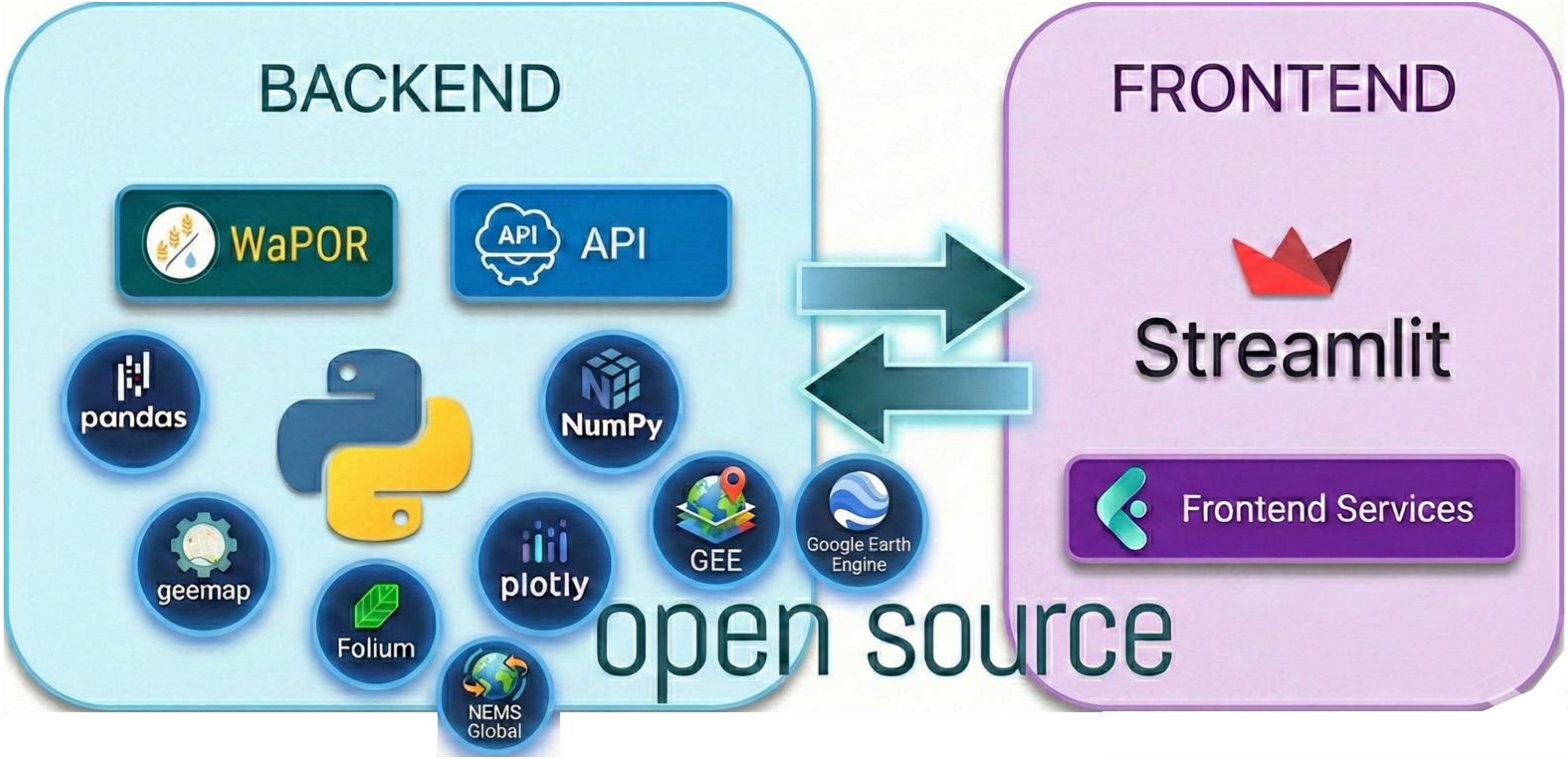
Key Features - Crop Intelligence



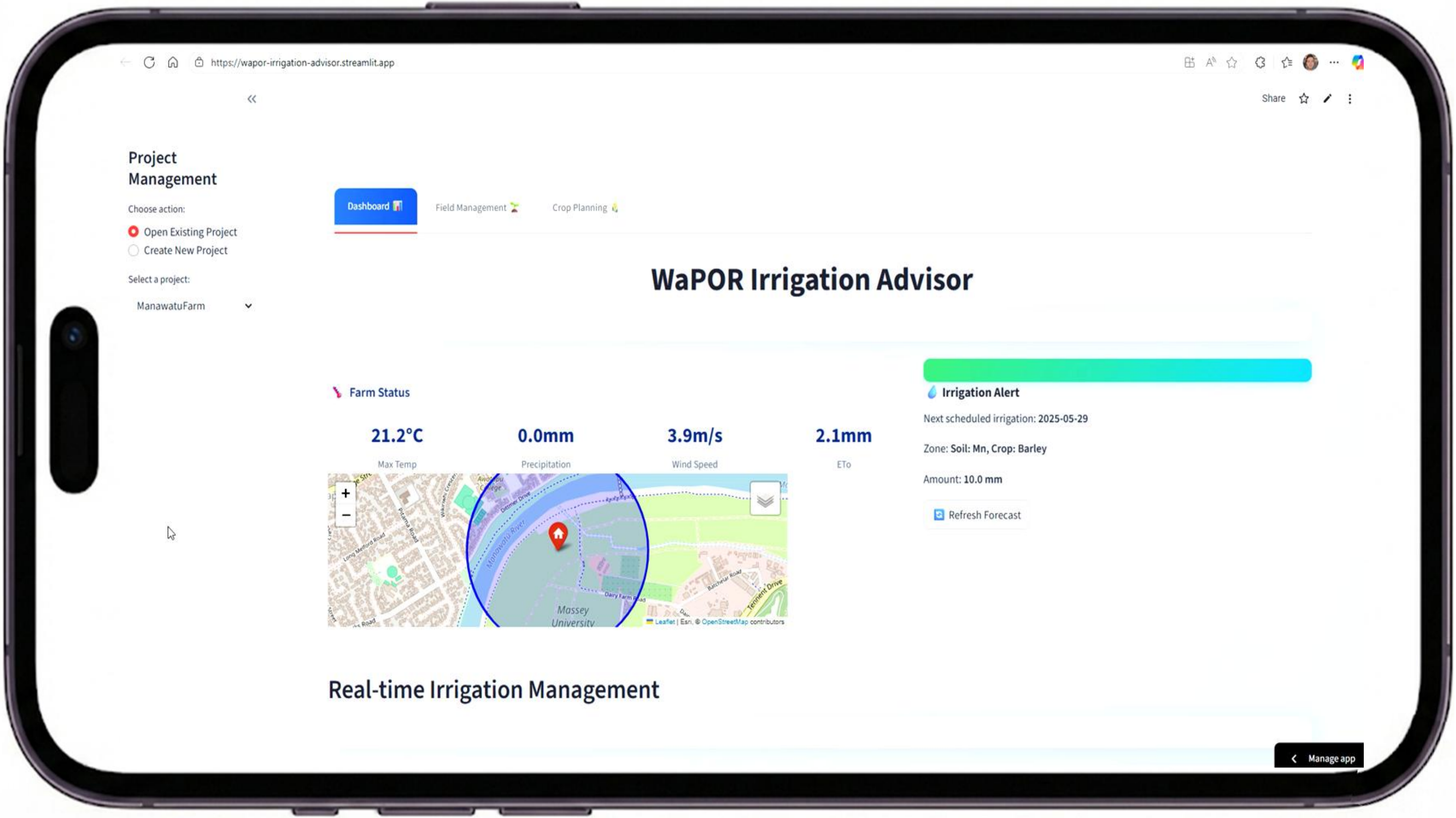
The Engine - Spatiotemporal Calculation



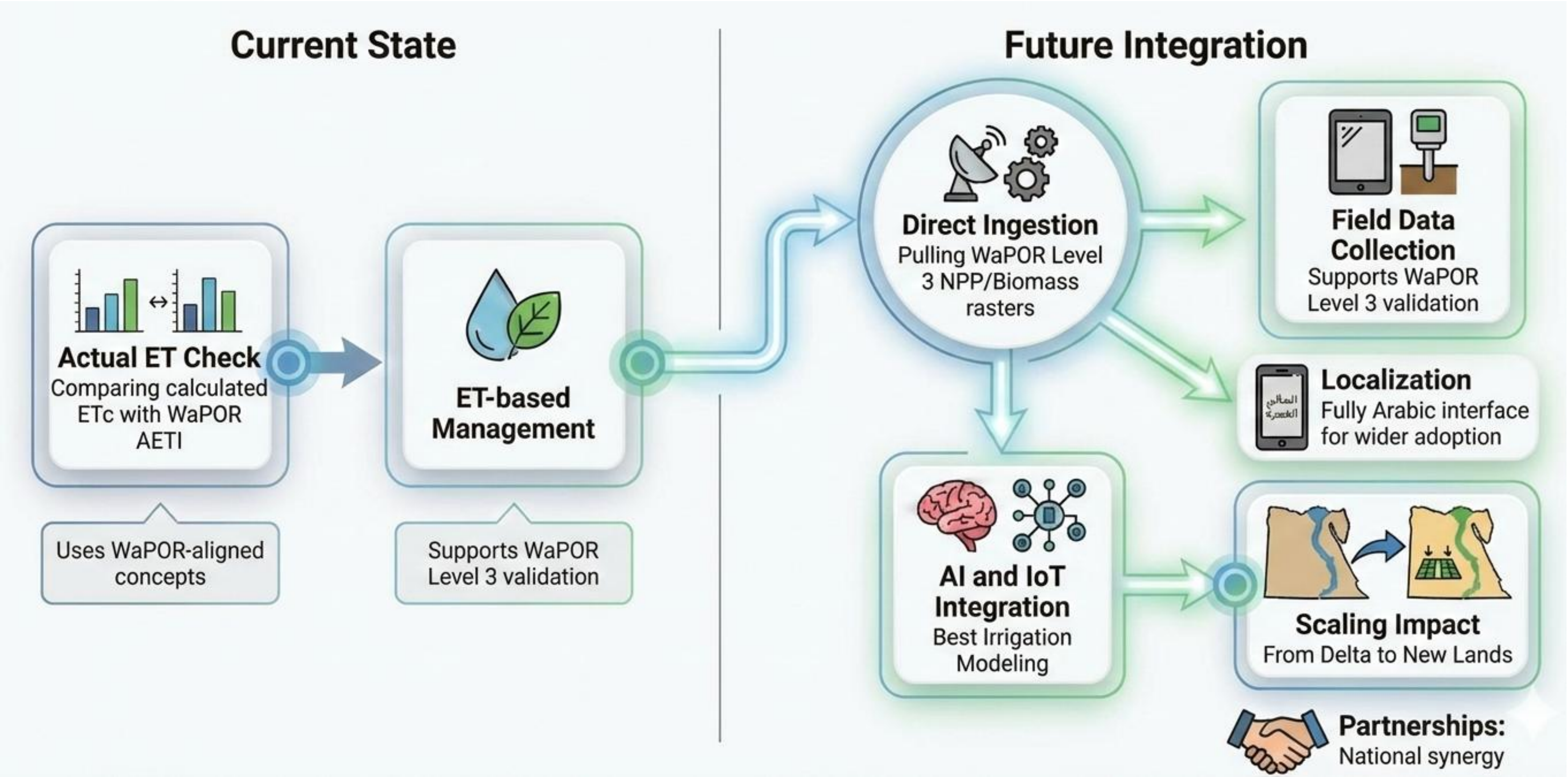
Technical Implementation




Live Demo /
Walkthrough
START DEMO
SELECT SCENARIO
VIEW ANALYTICS
SETTINGS
Step 1 of 5


https://wapor-irrigation-advisor.streamlit.app
Share
Project Management
Choose action:
Open Existing Project
Create New Project
Select a project:
ManawatuFarm
Dashboard
Field Management
Crop Planning
WaPOR Irrigation Advisor
Farm Status
21.2°C
Max Temp
0.0mm
Precipitation
3.9m/s
Wind Speed
2.1mm
ETo
Irrigation Alert
Next scheduled irrigation: 2025-05-29
Zone: Soil: Mn, Crop: Barley
Amount: 10.0 mm
Refresh Forecast
Manawatu River
Massey University
Dairy Farm Road
Batchelar Road
Tennent Drive
Long Melford Road
Leaflet | Esri, © OpenStreetMap contributors
Real-time Irrigation Management
Manage app

Future Outlook



Conclusion & Foreseen Impact

Summary

The **WaPOR Irrigation Advisor bridges the gap between high-level satellite data and farm-level execution.** Empowers stakeholders with data-driven decisions.

Empowers stakeholders with **data-driven decisions.**

Impact

Potential for **10-20% water saving** by avoiding over-irrigation.

Better yield stability through precise timing.

Call to Action

Collaboration with **Egyptian partners** (Ministry, WUAs) for field testing.

Way Forward & Collaboration